

AERYN DUNMORE

RESUMÉ | CURRICULUM VITAE

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Previously : Erin Chapman

I am a researcher in Machine Learning and Emotion Detection, currently a fully-funded PhD student at The University of Auckland's Bioengineering Institute, under the supervision of Professor Mark Billingham. My research focuses on the improvement of multimodal emotion recognition for conversational agents to improve mental health outcomes, with the first paper focusing on improving speech recognition with a new optimisation algorithm for machine learning. My thesis research began at Massey University, and was transferred to Auckland at the start of 2025. I have strong interests in the areas of cybersecurity, privacy, and ethics in STEM research, as well as my current focus in Machine Learning. I have an ongoing interest in cryptographic design. I created and implemented a symmetric encryption scheme for my Master's thesis, which led to the publication of two articles : a conference paper and the cover story for Digital Forensics Magazine. I have also worked with the Radio Astronomy Lab at AUT on schemes to generate large volumes of True Random Numbers using radio telescopes, which also became a conference paper. I began my Doctoral study at the University of Oxford, becoming seriously ill during the end of my first year. Because of this development leaving me disabled, I had to withdraw. I am still on good terms with the team who ran the Centre for Doctoral Training in Cybersecurity at Oxford, and the previous program head Professor Andrew Martin can provide an academic reference. My current research is funded by the Ministry of Business, Innovation, and Employment, in partnership with Soul Machines. My portion of the project research aims to identify user emotions over time, with a passive recognition model using biological signals and smart phone data, and in real-time for active use with additional audio and video.

CERTIFICATIONS

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| 2025-present | Doctor of Philosophy, (PhD) , (Bioengineering) Auckland Bioengineering Institute, The University of Auckland
Thesis : <i>Real-Time Multimodal Emotion Recognition for Low-Resource Devices</i>
Supervisors : Professor Mark Billingham, Dr Kunal Gupta, Dr Elliott Wen, Professor Ruili Wang |
| 2023-2025 | Doctor of Philosophy, (PhD) , (Computer Science) College of Sciences, Massey University
Thesis research transferred to The University of Auckland Thesis : <i>Real-Time Multimodal Emotion Recognition for Low-Resource Devices</i>
Supervisors : Professor Ruili Wang, Professor Mark Billingham, Dr Feng Hou |
| 2017-2018 | Doctor of Philosophy (DPhil) , Centre for Doctoral Training in Cybersecurity, Pembroke College, University of Oxford
(Incomplete due to illness) <ul style="list-style-type: none">> Created a newsletter for the Department of Cyber Security events - (in)Security.> Attended the InterACE Hackathon at The University of Cambridge as part of the Oxford Competitive Computer Society.> Reached the semi-final round in the Atlantic Council Cyber 9/12 Strategy Challenge, in a team for the University of Oxford, at BT Tower, London |
| 2015-2016 | Master of Computer and Information Sciences (Honours, 1st Class) , Auckland University of Technology.
Thesis Title : <i>Using Graphic Based Systems to Improve Cryptographic Algorithms</i> <ul style="list-style-type: none">> Invited Talk : 22nd July 2016
"Advancing stream cipher performance and computability", <i>Winter Research Series 2016</i>, Digital Forensics Laboratory, Auckland University of Technology> Dean's Award for Excellence in Postgraduate Study |
| 2011-2014 | Bachelor of Science (Computer Science) , University of Auckland |

WORKS IN PROGRESS

- 2026 **Dunmore, A.**, Billingham, M., Gupta, K., Wen, E., Wang, R, *Speech Emotion Recognition using Variable Length Speech Data Through New Vectorization Approaches*

PAPERS UNDER REVIEW

- 2026 **Dunmore, A.**, Billingham, M., Gupta, K., Wen, E., Nassani, A., Wang, R, *Efficient Optimisation of Hyper-Parameters for Increases in Speech Emotion Recognition Accuracy*, IEEE Transactions on Affective Computing

PUBLISHED

- 2023 **Dunmore, A.**, Jang-Jaccard, J., Sabrina, F., and Kwak, J., *MAGNETO and DeepInsight: Extended Image Translation with Semantic Relationships for Classifying Attack Data with Convolutional Neural Networks*, Electronics, August 2023
- 2023 **Dunmore, A.**, Samandari, J., and Jang-Jaccard, J., *Matrix Graph Walks for Lightweight Encryption*, Cryptography, August 2023
- 2023 **Dunmore, A.** and Jang-Jacquard, J., Sabrina, F., and Kwak, J., *A Comprehensive Survey on Generative Adversarial Networks (GANs) in Cybersecurity Intrusion Detection*, IEEE Access, August 2023
- 2023 Wei, Y., Jang-Jaccard, J., Sabrina, F., Xu, W., Camtepe, S., and **Dunmore, A.** *Reconstruction-based LSTM-Autoencoder for Anomaly-based DDoS Attack Detection over Multivariate Time-Series Data*, April 2023
- 2023 Zhu, J., Jang-Jaccard, J., Welch, I., Al Sahaf, H., Camtepe, S., and **Dunmore, A.** *Relation-Aware Based Siamese Denoising Autoencoder for Malware Few-Shot Classification*, SSRN Electronic Journal, January 2023
- 2016 Cusack, B., **Chapman, E.** *Challenging Cryptographic Performance*. Digital Forensics Magazine, Issue 29, November 2016
- 2016 **Chapman, E.**, Grewar, J., Nautsch, T. *Celestial Sources for Random Number Generation*. 14th Australian Information Security Management Conference, December 5-6, 2016
- 2016 Cusack, B., **Chapman, E.** *Using Graphic Methods to Challenge Cryptographic Performance*. 14th Australian Information Security Management Conference, December 5-6, 2016

PROJECTS

TÖKU HOA : A PERSONALISED AGENT FOR MENTAL HEALTH

2023-PRESENT

 [Project Site at MBIE.govt.nz](#)  [Project Lead : Professor Mark Billingham](#)

This project, with funding from MBIE's Catalyst Strategic Investment Fund, in partnership with Soul Machines, aims to develop a smartphone application with a virtual avatar, which monitors a user's emotional state using a combination of audio, visual, and biological signals.

Emotion Detection Machine Learning Interactive Applications Mental Health Multimodal Signals

CYBER SECURITY RESEARCH PROGRAMME - ARTIFICIAL INTELLIGENCE FOR AUTOMATING RESPONSE TO THREATS

2022-2023

 [Project Site at MBIE.govt.nz](#)  [Project Lead : Julian Jang-Jacquard](#)

This project was one of three projects underway as part of the Ministry of Business, Innovation, and Employment's *Catalyst : Strategic – The Cyber Security Research Programme*. The focus was on using new machine learning techniques to combat increases in the threat landscape, particularly malware and intrusion detection. These three projects funded by MBIE were being undertaken in conjunction with Australian collaborators, and coordinated by CSIRO's Data61 Group. My role involved exploring Generative Adversarial Networks for this purpose. Associate-Professor Jang-Jacquard and I finished two survey papers in this area for submission, as well as an original research paper into the use of context-aware image translation for network data to train machine learning models to detect intrusions and attacks.

Machine Learning Generative Adversarial Networks Malware Intrusion Detection Systems AI Python Microsoft Malware Challenge Contagio Drebin VirusShare VirusTotal

★ HONOURS/AWARDS

- 2018 **Award for Best Written Brief, Team Award : Oxford University**, 2018 Cyber 9/12 Student Challenge Conference, Scrowcroft Center for Strategy and Security, Atlantic Council, London
- 2017 **Recipient of the Dean's Award for Excellence in Postgraduate Study**, Auckland University of Technology.
- 2016 **Invited Talk : "Advancing stream cipher performance and computability"**, Winter Research Series 2016, Digital Forensics Laboratory, Auckland University of Technology, 22nd July 2016

SKILLS

Programming	Machine Learning, Neural Networks, Python, Symmetric cryptography, Java, C, C++, MatLab, Weka, SQL
Development Environments	Kaggle, Jupyter Notebooks, CodeRunner
Systems	Mac OS X, Windows, Linux
General Skills	LaTeX, Overleaf, Word, Excel, PowerPoint, Pages, Numbers, Academic writing, Communication, Web Design

EXPERIENCE

July 2023 June 2022	Research Support Assistant, MASSEY UNIVERSITY, Albany > Development of two survey papers into the state of the art in Generative Adversarial Networks in Malware and Intrusion Detection Systems. > Implementation of Python code for multiple machine learning techniques including GANs and auto-encoders. Python Machine Learning Neural Networks GAN Autoencoder Microsoft Malware Challenge 2015 Dataset Collaborative Research Academic writing
April 2022 September 2014	Technical Support, Website Maintenance, and Database Technician, UPSTART PRESS LTD, Auckland <i>IT support and website maintenance, part-time until 2020, then full-time until 2022.</i> > Managing the company's online sales website Upstart Press > Creation and maintenance of the Upstart Distribution website for updating the price and availability of products for Trade customers. > Database maintenance and website rebuild for Michael Cooper - an online subscription service for independent wine reviews. Python PHP SQL HTML Wordpress
2015 2012	Retail Sales Assistant, WHITCOULLS SYLVIA PARK, Auckland General sales assistant and customer queries. > Made manager of the Magazine Department 2014/2015. People Skills Scheduling Administration Customer Service
2012 2010	Fixed Line and Broadband Customer Service Representative. , VODAFONE NZ LTD, Auckland > Technical support for any and all queries related to fixed line, broadband, wireless routers, and computer connectivity. > Billing agent dealing with queries about financial documents from and for Vodafone Fixed-Line and Broadband Highlights : > Creation of a new testing program for Mac OS devices, Vodafone Diagnostics, for agent use in diagnosing speed issues. This program was subsequently rolled out company-wide. > Presenting a prior customer interaction to new support agents at company request to explain de-escalation techniques and the importance of customer follow up. > Numerous, highly positive customer reviews, including mention on a customer's blog. End of Year review document including customer feedback available on request. Fault Diagnosis Technical Communication EXE file creation Javascript Java Terminal and Console Use AppleScript

PROGRAMMING LANGUAGES

Python	● ● ● ● ●
Java	● ● ● ● ●
Javascript	● ● ● ● ○
C/C++	● ● ● ○ ○
PHP	● ● ● ○ ○

LANGUAGES

English (native)	● ● ● ● ●
French	● ● ○ ○ ○
Korean	● ○ ○ ○ ○
Russian	● ○ ○ ○ ○

INTERESTS

Cybersecurity ethics and responsibilities in STEM research, women in STEM, accessibility online, reading, writing, history, human rights, politics, philosophy, mathematical theory, classics.

OUTREACH AND VOLUNTEERING

2017	Invited speaker at <i>Girls' Innov8</i> , a bootcamp for learning to code at St Cuthbert's College, Auckland
2016-2017	Judge for the NIWA Science and Technology Fair, Annual, Auckland Region.
2008-2011	Costumed Guide for Howick Historical Village, Auckland.
2005-2015	Red Cross Annual Appeal volunteer.

REFERENCES

Professor Mark Billingham

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